

Dielectrics

1. Define polarization, dipole and dipole moment. Derive Clausius- Mossotti equation. **8 marks.**
2. What is internal field/Lorentz field? Derive Clausius Mossoti equation. **8 marks**
3. Explain polar and non-polar dielectrics. **5 marks**
4. Explain different electrical polarization mechanisms. **6 marks**
5. Describe solid, liquid and gaseous dielectrics with one example each. **6 marks**
6. Explain the application of dielectrics in transformer, capacitors and insulators. **6 marks**
7. An elemental solid dielectric material has a polarizability of $7 \times 10^{-40} \text{ F/m}^2$. Assuming the internal to be Lorentz field, calculate the dielectric constant for the material, if it has $3 \times 10^{28} \text{ atoms/ m}^3$.
(Ans: 12.28)
8. The dielectric constant of sulphur is 3.4. Assuming a cubical lattice for its structure, calculate the electronic polarisability of sulphur. Density of sulphur is 2.07 gm/cc, atomic weight is 32.07.
(Ans: $3.031 \times 10^{-40} \text{ Fm}^2$)
9. The dielectric constant of He gas at NTP is 1.0000684, Calculate the electronic polarisability of He atoms if the gas contains $2.7 \times 10^{25} \text{ atoms/ m}^3$. (Ans: $2.242 \times 10^{-41} \text{ Fm}^2$).
10. What is the polarization produced in sodium chloride by an electric field of 500 V/m. if it has relative permittivity of 6? (Ans: $2.214 \times 10^{-8} \text{ C/m}^2$)

Superconductivity

1. Describe Meissner's effect. Explain Types of superconductor. **8 marks**
2. Lead has superconducting transition temperature of 7.26 K. If initial field at 0K is $50 \times 10^3 \text{ A/m}$, calculate the critical field at 6K. **4 marks**
3. Explain the phenomenon of superconductivity. Discuss qualitatively the BCS theory of superconductivity. **8 marks**
4. Define superconductivity, critical temperature and critical field. **6 marks**
5. Explain the temperature dependence of critical field with suitable diagram. **6 marks**
6. Write a short note on high temperature superconductors. **4 marks**
7. Explain briefly SQUID and its application. **6 marks** [Note: mention Josephson effect with suitable diagram and write down about DC and RF Squids with suitable diagram]
8. Explain the construction and working of MAGLEV vehicle. **6 marks**

Lasers

1. Expand the acronym LASER and explain its characteristics. **5 marks**
2. Explain the interaction of radiation with matter. **6 marks** (note: with suitable diagram)
3. Derive an expression for energy density during radiation- matter interaction at thermal equilibrium and explain the relation between the Einstein coefficients. **8 marks**
4. Explain the requisites and condition to achieve Laser action. **5 marks**
5. Explain the construction and working of a CO₂ laser. **8 marks**
6. Explain the applications of Laser. **6 marks**
7. The ratio of population of two energy levels is 1.059×10^{-30} . Find the wavelength of the light emitted at 330 K. (Ans: 632 nm)
8. A laser emits radiation of wavelength $6328 \text{ \AA}$. And with an average power of 1.5 mW. Find the number of photons emitted per second by the laser. (Ans: $4.76 \times 10^{15}/\text{s}$).
9. Calculate the ratio of population in two energy levels between which the transition of electron is responsible for emission of wavelength of light of 684.2 nm at 300 K. (Ans: 9.39×10^{-31})